

University IT Asset Management System

Submission: In class

Deadline: 5/5/2026

Project Description

You need to develop a decentralized application that tracks university IT equipment (laptops, projectors, etc.), including:

- `device_type` (e.g., Laptop, Monitor, Projector)
- `brand`
- `purchase_year`
- `department`
- `assigned_to` (Name of the faculty/staff member)

Each IT asset will be identified with a unique ID (e.g., ASSET001).

Required Features

- **Create:** Users can add new IT assets to the ledger.
- **Read All:** Users can view all registered IT assets in the university.
- **Update:** Users can update the `assigned_to` information (e.g., transferring a laptop from one faculty member to another).
- **Search / Filter:** Users can search for assets by:
 1. Unique ID
 2. Department name (e.g., find all assets in "CSE")
 3. Device Type (e.g., find all "Projectors" across the entire university)

Instructions

You **do not need to modify** or manage:

- Wallet
- Admin
- User configuration

Do not change network configuration, `enrollAdmin.js`, or `registerUser.js`. Use these files as-is, similar to FabCar Lab.

For each transaction, you may reuse the existing code segments that gather:

- Chaincode configuration path
- Wallet data
- Network and contract instances

To further your development, here is a diagram that shows the flow of queries (getting data from API/chaincode) and the search feature based on our FabCar lab.

